EC5.407 Wireless Communications
Assignment 2: Detection and Diversity in AWGN and Fading Channels

Anumay Rai
Roll No. 2025122013

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Setup used in all problems

All simulations follow the Getting Started section. Constellations are scaled to unit average energy, so BPSK is $\{\pm 1\}$ and 4-QAM is $\{(\pm 1 \pm j)/\sqrt{2}\}$. For a given SNR in dB, the noise is $w \sim \mathcal{CN}(0, 1/\text{SNR})$ and Rayleigh coefficients are $h \sim \mathcal{CN}(0, 1)$, both generated with `standard_normal`. A fresh h and w is drawn for every symbol. The detector is one function that, given y, the effective gain g and the constellation, builds the $N \times M$ matrix of distances $|y_n - g_n u_m|^2$ and takes the argmin over m. The same detector is used for BPSK, 4-QAM and for every combiner in Problems 2 and 3 (each combiner only produces a scalar $z = gx + \tilde{w}$). Errors are counted by comparing indices. Every SNR point uses $N = 10^6$ symbols, and points with fewer than 10 errors are not plotted. The seeds are listed in the README.

1 Problem 1: SER in AWGN and Rayleigh fading

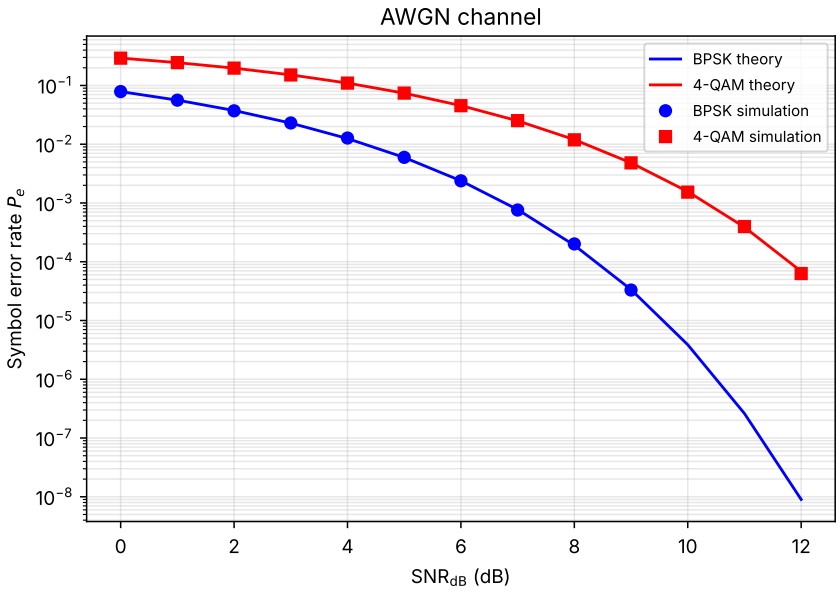


Figure 1: SER of BPSK and 4-QAM in the AWGN channel ($h = 1$). Markers: simulation, lines: theory. $\text{SNR}_{\text{dB}} = 0, 1, \ldots, 12$, $N = 10^6$ symbols per SNR point.

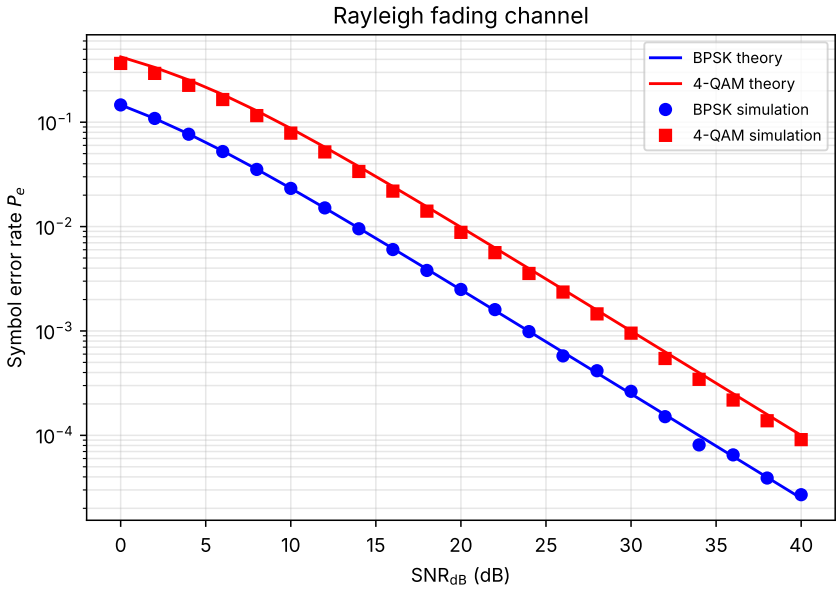


Figure 2: SER of BPSK and 4-QAM in the Rayleigh fading channel with coherent ML detection. Markers: simulation, lines: theory. $\mathrm{SNR_{dB}} = 0, 2, \ldots, 40$, $N = 10^6$ symbols per SNR point.

Agreement between simulation and theory

In AWGN (Fig. 1) the simulated points sit on the theoretical curves for both schemes. The relative difference is within a few percent wherever there are many errors, and grows to about 5–10% at the last points, which is what we expect since those points have only around 30–60 errors ($1/\sqrt{n_{\mathrm{err}}}$). For BPSK the simulation gives no errors at 10, 11 and 12 dB. This is not a bug: the true P_e there is 3.9×10^{-6} and lower, so 10^6 symbols are expected to give about 4 errors or less, and these points are dropped by the $n_{\mathrm{err}} \geq 10$ rule.

In Rayleigh fading (Fig. 2) BPSK matches the exact expression over the full range. For 4-QAM the theory curve lies slightly above the simulation at low SNR (0.423 vs. 0.366 at 0 dB) and the two merge as SNR increases. The reason is that the 4-QAM fading expression is only an approximation: it is $2P_b$, which double counts the event where both the I and Q components are wrong. At low SNR this event is not rare, so the approximation overestimates P_e. At high SNR this joint event becomes negligible and the curves coincide. So this deviation is from the formula, not the simulation.

AWGN vs. fading at high SNR

The two figures look very different. In AWGN the SER falls like $e^{-\mathrm{SNR}}$ (a "waterfall"), while in fading it falls only as $1/\mathrm{SNR}$: for BPSK, $P_e \approx 1/(4\,\mathrm{SNR})$, which is one decade per 10 dB. The measured slope between 30 and 40 dB is 0.99 for BPSK and 1.02 for 4-QAM. As an example, 4-QAM needs about 10.3 dB for $P_e = 10^{-3}$ in AWGN but about 29.8 dB in fading, a penalty of nearly 20 dB.

This is explained by the deep-fade argument. Given h, the error probability is $Q(\sqrt{2|h|^2\mathrm{SNR}})$, which is tiny when $|h|^2\mathrm{SNR} \gg 1$ and close to 1/2 when $|h|^2\mathrm{SNR} \lesssim 1$. So at high SNR almost all errors come from the channel being in a deep fade, $|h|^2 < 1/\mathrm{SNR}$. Since $|h|^2$ is exponential with mean 1, $\Pr(|h|^2 < 1/\mathrm{SNR}) = 1 - e^{-1/\mathrm{SNR}} \approx 1/\mathrm{SNR}$. The error rate is therefore set by how often the channel is bad, not by the noise, and it decays only linearly with SNR.

2 Problem 2: Receive diversity (SIMO)

Decision statistics

The received vector is $\mathbf{y} = \mathbf{h}x + \mathbf{w}$ with $\mathbf{w} \sim \mathcal{CN}(\mathbf{0}, \frac{1}{\mathrm{SNR}}\mathbf{I}_L)$.

MRC. I combine with the normalized weights $\mathbf{h}/\|\mathbf{h}\|$:

$$z_{\mathrm{MRC}} = \frac{\mathbf{h}^H\mathbf{y}}{\|\mathbf{h}\|} = \|\mathbf{h}\|\, x + \tilde{w}, \qquad \tilde{w} = \frac{\mathbf{h}^H\mathbf{w}}{\|\mathbf{h}\|} \sim \mathcal{CN}\big(0, \tfrac{1}{\mathrm{SNR}}\big)\,.$$

So MRC is a scalar channel with gain $\|\mathbf{h}\|$ and noise variance $1/\mathrm{SNR}$, i.e. an instantaneous SNR of $\|\mathbf{h}\|^2\mathrm{SNR}$. The detector of Problem 1 is called with $g = \|\mathbf{h}\|$. This is the same as the full vector ML rule, because

$$\|\mathbf{y} - \mathbf{h}u\|^2 = \|\mathbf{y}\|^2 - 2\,\mathrm{Re}\{u^*\mathbf{h}^H\mathbf{y}\} + |u|^2\|\mathbf{h}\|^2 = |z_{\mathrm{MRC}} - \|\mathbf{h}\|u|^2 + \Big(\|\mathbf{y}\|^2 - |z_{\mathrm{MRC}}|^2\Big),$$

and the last bracket does not depend on u. (If one does not normalize and uses $\mathbf{h}^H\mathbf{y} = \|\mathbf{h}\|^2 x + \mathbf{h}^H\mathbf{w}$, the distances to $\|\mathbf{h}\|^2 u$ are just the above multiplied by $\|\mathbf{h}\|^2 > 0$, so the argmin is the same.)

SC. With $k = \arg\max_\ell |h_\ell|^2$,

$$z_{\mathrm{SC}} = y_k = h_k x + w_k, \qquad w_k \sim \mathcal{CN}\big(0, \tfrac{1}{\mathrm{SNR}}\big)\,,$$

a scalar channel with gain h_k, noise variance $1/\mathrm{SNR}$ and instantaneous SNR $\max_\ell |h_\ell|^2\,\mathrm{SNR}$. The detector is called with $g = h_k$.

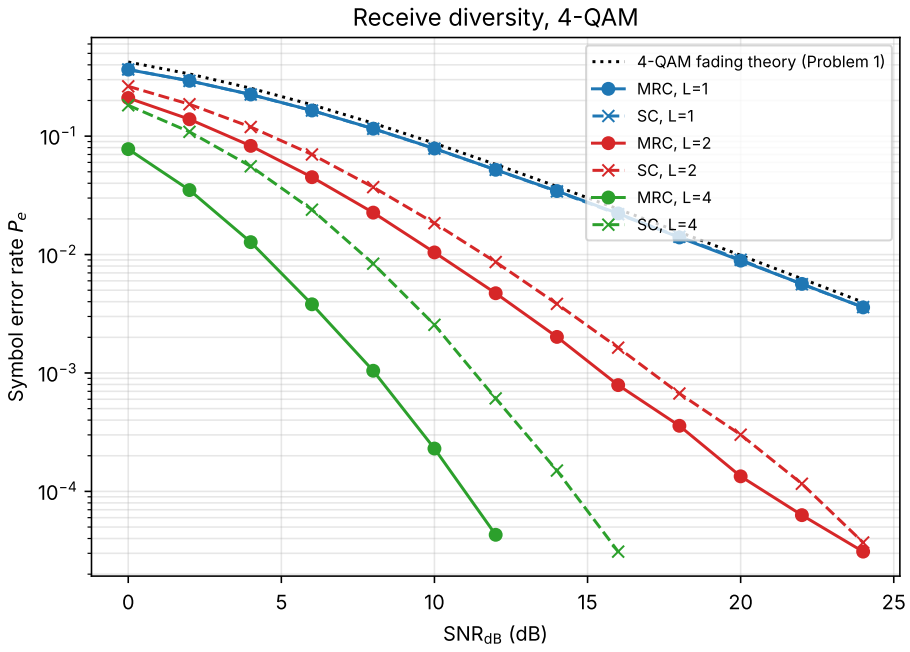


Figure 3: SER of 4-QAM with MRC (solid) and SC (dashed) for $L = 1, 2, 4$. Same colour for the same L. The dotted black line is the 4-QAM Rayleigh expression from Problem 1. $\mathrm{SNR}_{\mathrm{dB}} = 0, 2, \ldots, 24$ (per receive antenna), $N = 10^6$ symbols per SNR point.

Consistency check

For $L = 1$ both combiners reduce to $z = hx + w$, and the MRC and SC curves fall on top of each other (they differ only by Monte Carlo noise, e.g. 3.58×10^{-3} vs. 3.60×10^{-3} at 24 dB). They also match the 4-QAM fading result of Problem 1 (same small offset from the approximate formula at low SNR as before).

Slopes and diversity order

Both MRC and SC achieve diversity order L, i.e. $P_e \propto \mathrm{SNR}^{-L}$ at high SNR. The measured slopes (decades per 10 dB, taken over the last points with enough errors) are:

	$L=1$	$L=2$	$L=4$
MRC	0.99	1.88	3.24
SC	0.98	1.90	3.19
Predicted	1	2	4

$L=1$ and $L=2$ are close to the prediction. For $L=4$ the slope is still increasing at the end of the curve: the SER reaches 10^{-5} already at around 12–16 dB, so with 10^6 symbols we cannot go far enough into the high-SNR region to see the full slope of 4. Between consecutive points the local slope keeps getting steeper (for MRC it goes from about 2.2 between 2 and 4 dB to about 3.6 between 10 and 12 dB), which is consistent with an asymptotic order of 4.

Gap between MRC and SC

At $P_e = 10^{-3}$ SC needs about 1.6 dB more than MRC for $L=2$ and about 3.3 dB more for $L=4$ (similar numbers at 10^{-2}). Both schemes have the same diversity order, so the curves are parallel and the gap comes from the array gain, i.e. the average effective SNR. For MRC it is $\mathbb{E}[\|\mathbf{h}\|^2] = L$, while for SC it is $\mathbb{E}[\max_\ell |h_\ell|^2] = \sum_{k=1}^{L} 1/k$, which is 1.5 for $L=2$ and 2.08 for $L=4$. The ratio is $10\log_{10}(2/1.5) = 1.25$ dB for $L=2$ and $10\log_{10}(4/2.08) = 2.83$ dB for $L=4$. The simulated gaps are a bit bigger than these numbers since the SER depends on the whole distribution of the effective SNR and not only on its mean, but the trend is the same: SC throws away the energy from $L-1$ antennas, and this loss grows with L.

3 Problem 3: Transmit diversity (MISO, $L=2$)

The total transmitted energy per channel use is kept at 1 in all schemes, and the noise is $\mathcal{CN}(0, 1/\mathrm{SNR})$ as before.

Single antenna. $y = hx + w$, same as Problem 1.

Transmit beamforming. The transmitter knows $\mathbf{h} = (h_1, h_2)^T$ and sends $s_\ell = v_\ell x$ with

$$\mathbf{v} = \frac{\mathbf{h}^*}{\|\mathbf{h}\|}, \qquad y = h_1 v_1 x + h_2 v_2 x + w = \|\mathbf{h}\|\, x + w.$$

The two components add in phase. Energy per channel use is $|v_1|^2 + |v_2|^2 = 1$ times $|x|^2$, so the scaling is $1/\|\mathbf{h}\|$. The decision statistic is y itself, with gain $\|\mathbf{h}\|$ and noise variance $1/\mathrm{SNR}$, exactly the same scalar channel as MRC with $L=2$.

Alamouti. Two symbols x_1, x_2 are sent over two channel uses (rows: time, columns: antennas):

$$\mathbf{S} = \frac{1}{\sqrt{2}} \begin{pmatrix} x_1 & x_2 \\ -x_2^* & x_1^* \end{pmatrix}, \qquad \begin{aligned} y_1 &= \tfrac{1}{\sqrt{2}}(h_1 x_1 + h_2 x_2) + w_1, \\ y_2 &= \tfrac{1}{\sqrt{2}}(-h_1 x_2^* + h_2 x_1^*) + w_2. \end{aligned}$$

The factor $1/\sqrt{2}$ is the scaling: each antenna gets half the energy, so the energy per channel use is $(|x_1|^2 + |x_2|^2)/2$, which is 1 on average. The receiver forms

$$z_1 = \frac{h_1^* y_1 + h_2 y_2^*}{\|\mathbf{h}\|} = \frac{\|\mathbf{h}\|}{\sqrt{2}} x_1 + \tilde{w}_1, \qquad z_2 = \frac{h_2^* y_1 - h_1 y_2^*}{\|\mathbf{h}\|} = \frac{\|\mathbf{h}\|}{\sqrt{2}} x_2 + \tilde{w}_2,$$

with $\tilde{w}_1, \tilde{w}_2 \sim \mathcal{CN}(0, 1/\mathrm{SNR})$ and independent. Each symbol sees a scalar channel with gain $\|\mathbf{h}\|/\sqrt{2}$, so the detector of Problem 1 is called with that gain. $\mathbf{h}$ is drawn once per block and the SER is counted over all N symbols ($N/2$ blocks).

Energy check. The average transmitted energy per channel use measured in the simulation (averaged over all SNR points) is 1.000000 for the single antenna, beamforming and Alamouti schemes, so the comparison is fair.

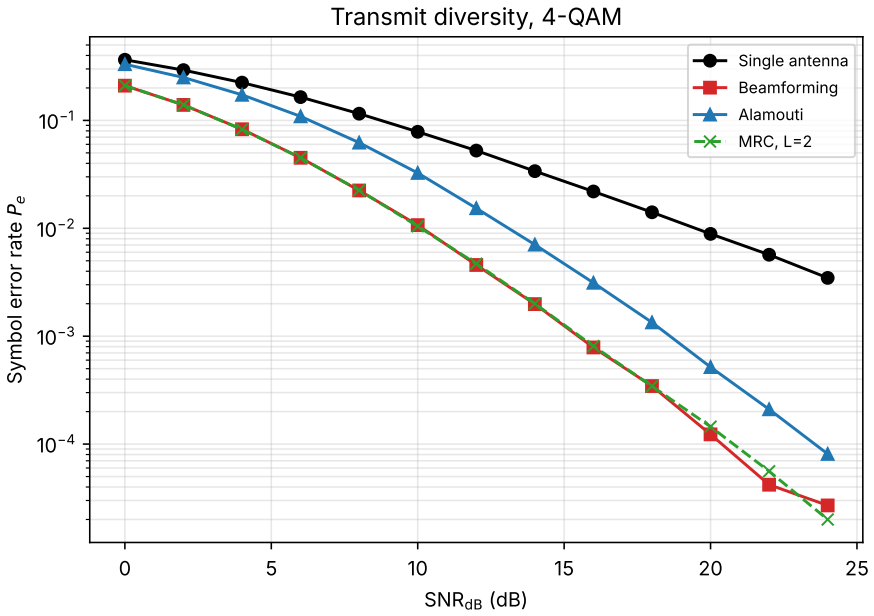


Figure 4: SER of 4-QAM with a single antenna, transmit beamforming and Alamouti ($L = 2$ transmit antennas), and receive MRC with $L = 2$ from Problem 2. $\mathrm{SNR_{dB}} = 0, 2, \ldots, 24$ (total transmit energy normalized to 1), $N = 10^6$ symbols per SNR point.

Diversity order

Measured slopes between 12 and 22 dB: single antenna 0.97, beamforming 2.04, Alamouti 1.86, MRC ($L = 2$) 1.92. So the single antenna has diversity 1 and all three two-antenna schemes have diversity 2. The curves for the two-antenna schemes are parallel.

SNR gaps

Beamforming and MRC with $L = 2$ are on top of each other (within about 0.1 dB, which is Monte Carlo noise). This is expected since both give $\|\mathbf{h}\|x + w$ with the same noise variance: one does the coherent combining at the transmitter, the other at the receiver.

Alamouti is about 3 dB worse: it needs 18.6 dB for $P_e = 10^{-3}$ against 15.5 dB for beamforming/MRC (13.1 vs. 10.1 dB at 10^{-2}). From the decision statistic, the effective SNR of Alamouti is $\|\mathbf{h}\|^2\mathrm{SNR}/2$, half of beamforming. Since the transmitter does not know the channel it cannot steer the signal, so it splits the power equally between the two antennas and the two signals do not add coherently at the receiver. It gets the full diversity gain but loses the array gain of $10\log_{10} 2 = 3$ dB. MRC does not have this problem because the two receive antennas each collect the full transmitted energy.

References

[1] R. Gangula, *Performance in Narrowband Fading Channels*, Lecture Notes, EC5.407, IIIT Hyderabad.

[2] R. Gangula, *Diversity in Narrowband Fading Channels*, Lecture Notes, EC5.407, IIIT Hyderabad.